

$$\nabla_\alpha V^\mu = \underbrace{\partial_\alpha V^\mu}_{\frac{\partial}{\partial x^\alpha}} + \underbrace{\Gamma_{\beta\alpha}^\mu}_{\frac{\partial e_{(\beta)}}{\partial x^\alpha} = \Gamma_{\beta\alpha}^\mu e_\mu} V^\beta = (V^\mu)_{;\alpha}$$

Pg 1

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad \text{in polar} \quad dr^2 + r^2 d\varphi^2 \Rightarrow g = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

$$\nabla_\alpha T^\mu{}_\nu{}^\rho = \partial_\alpha T^\mu{}_\nu{}^\rho + \Gamma_{\sigma\alpha}^\mu T^\sigma{}_\nu{}^\rho - \Gamma_{\nu\alpha}^\sigma T^\mu{}_\sigma{}^\rho + \Gamma_{\sigma\alpha}^\rho T^\mu{}_\nu{}^\sigma$$

$$x^\mu, x^{\mu'}, \quad \Lambda_{\mu}^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu}$$

Show that $\partial_\alpha V^\mu$ does not transform like a tensor

$$\begin{aligned} V^{\mu'}_{;\alpha'} &= \frac{\partial}{\partial x^{\alpha'}} V^{\mu'} = \frac{\partial x^\rho}{\partial x^{\alpha'}} \frac{\partial}{\partial x^\rho} \Lambda_{\nu}^{\mu'} V^\nu = \Lambda_{\alpha'}^\rho \frac{\partial \Lambda_{\nu}^{\mu'}}{\partial x^\rho} V^\nu + \Lambda_{\alpha'}^\rho \Lambda_{\nu}^{\mu'} \frac{\partial}{\partial x^\rho} V^\nu \\ &= \underbrace{\Lambda_{\alpha'}^\rho \Lambda_{\nu}^{\mu'} V^\nu_{;\rho}}_{\text{this is how a tensor would transform}} + \underbrace{\Lambda_{\alpha'}^\rho \Lambda_{\nu}^{\mu'}{}_{;\rho} V^\nu}_{\text{extra term}} \end{aligned}$$

this is how a tensor would transform

extra term

that captures how the metric changes

But the covariant derivative $\nabla_\alpha V^\mu$ transforms like a tensor.

Insist that

① $(\phi, \alpha)_{\beta} = (\phi, \beta)_{\alpha}$ $\rightarrow$ why would this make sense?
 $\hookrightarrow$ example?

$$\nabla_\beta \partial_\alpha \phi = \nabla_\alpha \partial_\beta \phi$$

$$\partial_\beta \partial_\alpha \phi - \Gamma_{\alpha\beta}^\nu \phi_{,\nu} = \phi_{,\beta\alpha} - \Gamma_{\beta\alpha}^\nu \phi_{,\nu}$$

~~$$\phi_{\alpha\beta} - \Gamma_{\alpha\beta}^{\nu} \phi_{,\nu} = \phi_{\beta\alpha} - \Gamma_{\beta\alpha}^{\nu} \phi_{,\nu}$$~~

$$\Leftrightarrow (I) \quad T_{\alpha\beta}^{\nu} = T_{\beta\alpha}^{\nu}$$

$$(2) \quad V_{\alpha i \beta} = (g_{\alpha \nu} V^{\nu})_{i \beta}$$

$$\text{inst} \quad || \quad = (g_{\alpha\nu})_{;\beta} V^\nu + \underline{g_{\alpha\nu}(V^\nu)_{;\beta}}$$

$$= \cancel{(g_{\alpha\beta})_{;\beta} V^\beta} + \cancel{g_{\alpha\beta} [V_{;\beta} + T^\mu_{\nu\beta} V^\mu]}$$

$$g_{\alpha\nu} (V^\nu)_{;\beta}$$

$$(II) \rightarrow (g_{\alpha\nu})_{;\beta} = 0$$

→ what does this mean?

$$(a) \quad g_{\alpha\beta;\mu} = g_{\alpha\beta,\mu} - \cancel{T_{\alpha\mu}^{\nu} g_{\nu\beta}} - \cancel{T_{\beta\mu}^{\nu} g_{\alpha\nu}} = 0$$

$$(b) \quad g_{\beta\mu;\alpha} = g_{\beta\mu,\alpha} - \Gamma_{\beta\alpha}^{\nu} g_{\nu\mu} - \cancel{\Gamma_{\mu\alpha}^{\nu} g_{\beta\nu}} = 0$$

$$(c) \quad g_{\mu\alpha;\beta} = g_{\mu\alpha,\beta} - T_{\mu\beta}^{\nu} g_{\nu\alpha} - T_{\alpha\beta}^{\nu} g_{\mu\nu} = 0$$

shuffle

$$g^{\mu\nu} = g^{\nu\mu}.$$

$$(a) - (b) - (c) = g_{\alpha\beta, \mu} - g_{\beta\mu, \alpha} - g_{\mu\alpha, \beta} + 2 T_{\alpha\beta}^{\nu} g_{\mu\nu} = 0$$

$$T^{\nu}_{\alpha\beta} g^{\mu\nu} g_{\mu\nu} = \frac{1}{2} g^{\mu\nu} (g_{\alpha\mu,\beta} + g_{\beta\mu,\alpha} - g_{\alpha\beta,\mu})$$

II. $\Gamma_{\alpha\beta}^{\gamma} = \frac{1}{2} g^{\mu\nu} (g_{\alpha\mu,\beta} + g_{\beta\mu,\alpha} - g_{\alpha\beta,\mu})$

What does this mean?

Physic

→ Cartesian space $\frac{d}{d\lambda} \equiv \frac{dx^\mu}{d\lambda} \frac{\partial}{\partial x^\mu} = 0$

think about $\frac{D}{d\lambda} = \frac{dx^\mu}{d\lambda} \nabla_\mu$ generalizing this:

trajectory
 $x^\mu(\lambda)$
 $\uparrow$
 τ (proper time).

 $\frac{DT}{d\lambda} \Rightarrow$ ~~covariantly~~ Constant along trajectory.

along its trajectory,
its velocity is constant $\frac{D}{d\lambda} V^\nu = \frac{dx^\mu}{d\lambda} (V^\nu_{;\mu} + \Gamma^\nu_{\sigma\mu} V^\sigma)$

No change in velocity

$$\boxed{\frac{D}{d\lambda} \frac{dx^\nu}{d\lambda} = 0} \text{ in flat space}$$

Generalization of
1st Newton's law

$$\frac{dx^\mu}{d\lambda} \left(\frac{\partial}{\partial x^\mu} \frac{dx^\nu}{d\lambda} + \Gamma^\nu_{\sigma\mu} \frac{dx^\sigma}{d\lambda} \right) = 0$$

Difference between

$$\frac{\partial}{\partial x^\mu} \text{ & } \frac{\partial}{\partial x^\mu}$$

why is this the der.

$$= \frac{d}{d\lambda} \frac{dx^\nu}{d\lambda} + \Gamma^\nu_{\sigma\mu} \frac{dx^\mu}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0$$

"velocity of an object is constant along its trajectory"

$$\boxed{\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\sigma\mu} \frac{dx^\sigma}{d\lambda} \frac{dx^\mu}{d\lambda} = 0}$$

the geodesic equation.

given $g_{\mu\nu}$, you'll know how particles travel

Einstein's Eqns

tells you $g_{\mu\nu}$ given mass/

How much of the metric is physical?

independent of coordinates? ②

$$g_{\mu\nu}^{(x)} = g_{\mu\nu}(x_0) + (x-x_0)^\sigma g_{\mu\nu,\sigma} \Big|_{x_0} + \frac{1}{2} (x-x_0)^\sigma (x-x_0)^\tau g_{\mu\nu,\sigma\tau} \Big|_{x_0} + \dots$$

$$g_{\mu'\nu'} = \Lambda_{\mu'}^\mu \Lambda_{\nu'}^\nu g_{\mu\nu}$$

$$\Lambda_{\mu'}^\mu(x) = \Lambda_{\mu'}^\mu(x_0) + (x-x_0)^\sigma \Lambda_{\mu',\sigma}^\mu + \dots$$

this is a very deep argument. Spend some more time on this.

symmetric		d-dimensions.	d=4
$\binom{d}{2} \rightarrow g_{\mu\nu}^{(x_0)}$		$\frac{d(d+1)}{2}$	10
$= \frac{d!}{(d-2)!2!} = \frac{d(d-1)}{2}$	$g_{\mu\nu,\sigma}$	$d \left(\frac{d(d+1)}{2} \right)$	40
② $\rightarrow g_{\mu\nu,\sigma\tau}$		$\left[\frac{d(d+1)}{2} \right]^2$	100
$\nwarrow \Lambda_{\mu'}^\mu$		d^2	16
$\nwarrow \Lambda_{\mu',\sigma}^\mu$		$d \left[\frac{d(d+1)}{2} \right]$	40
$\nwarrow \Lambda_{\mu',\sigma\tau}^\mu$		$d \left[\frac{d(d+1)(d+2)}{6} \right]$	80

$g_{\mu\nu} \rightarrow \eta_{\mu\nu}$

diagonalizing & redefining coordinates

6 transformations $\rightarrow$ 3 Lorentz boost

$\rightarrow$ 3 Rotations.

Curvature = what can't be removed by coordinate changes.

* Math toolkit
ind. components
- Counting symmetric & antisymmetric tensors.

20 independent components.

complete character - 2nd derivatives